

MEETING SUMMARY

Mentor Review — Agentic AI for Automotive Spare Parts Management

Project	Agentic AI System for Automotive Spare Parts Management
Meeting Type	Mentor Review
Date	6 Aug 2026 (Thu)
Purpose	Track implementation progress

1. Summary of Discussion

Reviewed current progress of the project. Following changes were agreed on for the next phase of work, covering the Vision Agent, a review step after extraction, the demo UI, and how shared parts across assemblies are handled.

1.1 Vision Agent

The Vision Agent is currently a prototype and needs to be fully integrated into the pipeline.

- Upgrade to a **stronger vision-language model** and validate it on multiple exploded-view diagrams.
- Perform proper evaluation across different assemblies.
- Extend the Vision Agent output to include the **required quantity of each part** along with the Visual BOM.

1.2 Human-in-the-Loop After Extraction

The current pipeline proceeds directly from extraction to inventory and report generation without verification. To improve reliability:

- Introduce a mandatory **Human-in-the-Loop** checkpoint after the extraction stage.
- Display the extracted BOM for review, allowing users to **edit, add, or remove** line items before approving the pipeline to continue.
- Explore about **automated cross-verification** step after extraction from Document Agent and Vision Agent.

1.3 Demo UI

Develop a simple interface (preferably **Streamlit**) for the final demonstration and demo video.

- Allow users to submit assembly requests and monitor the pipeline execution.
- Integrate the HITL stage by displaying the extracted BOM with editing and approval options.
- Present the final report, inventory status, procurement decisions, and recommendations within the same interface.

1.4 Shared Parts Across Assemblies

Enhance the Inventory/Decision Agent to identify parts shared across multiple assemblies.

- Check whether a part number appears in other assemblies.
- Flag shared parts in the final output to inform users about cross-assembly dependencies.
- This will improve procurement prioritization and provide better inventory planning by highlighting components with higher operational impact.

2. Architecture Reference

Pipeline shared with mentor for context:

